



11374 - A JWST/MIRI Search for Elements Beyond the Iron Peak in Cassiopeia A

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Charles Kilpatrick (PI)	Northwestern University
Dr. Dan Milisavljevic (CoI)	Purdue University
Dr. Tea Temim (CoI)	Princeton University
J. Martin Laming (CoI)	Naval Research Laboratory
Dr. Robert A. Fesen (CoI)	Dartmouth College
Prof. Ilse De Looze (CoI) (ESA Member)	Universiteit Gent
Dr. John Charles Raymond (CoI)	Smithsonian Institution Astrophysical Observatory
Ms. Danielle Dickinson (CoI)	Purdue University

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	MIRI IFU Ti44 1	MIRI Medium Resolution Spectroscopy	(2) CASSIOPEIA-A-TI44-1
	2	MIRI IFU Outflow 1	MIRI Medium Resolution Spectroscopy	(1) CASSIOPEIA-A-OUTFLOW-1
	3	MIRI IFU Background	MIRI Medium Resolution Spectroscopy	(3) BACKGROUND

ABSTRACT

Roughly half of the elements heavier than the iron peak are synthesized in regions with extremely high neutron fluxes, thought to be released from the degenerate cores of massive stars (i.e., neutron stars). Although neutron star mergers are known to produce some of these heavy elements, their rates, total yield, and abundance profiles remain wildly uncertain. Low-metallicity stars also require prompt cosmic heavy element production, pointing to other sites of rapid neutron star capture to make up the balance. Core-collapse supernovae (CCSN) may explain the "first peak" of heavy elements as CCSN rates are high, they produce neutron stars, and some of their remnants exhibit morphological evidence for nuclear enrichment

from neutron stars. In particular, Cassiopeia A (Cas A), the youngest core-collapse supernova remnant in our Galaxy, hosts a bipolar "jet" containing a small fraction of its unshocked ejecta and pointing back to its central neutron star. Recent JWST observations revealed a collimated, unshocked [O IV] outflow that may be associated with these ejecta. We propose to use the MIRI IFU and medium resolution spectroscopy to target two positions in the interior of Cas A "jet" - one at the brightest point of the [O IV] outflow and another further interior and coincident with iron-peak elements as traced by Ti-44 - in order to constrain the abundance of heavy elements such as germanium, selenium, and arsenic that trace neutron capture during core collapse. Combined these will be the deepest mid-infrared spectra obtained of Cas A and a legacy data set for studying nucleosynthesis, ejecta properties, and details of the Cas A explosion.

OBSERVING DESCRIPTION

This program aims to obtain extremely deep MIRI/MRS observations of carefully selected positions in Cassiopeia A that will probe forbidden lines from heavy elements in its unshocked ejecta. Due to the wavelength of the lines we are targeting, we request only MIRI/MRS MEDIUM bandpass observations, which will be dithered to optimally probe emission across the resolved components of ejecta in the interior of the remnant.

There are two sets of observations probing two sites along the eastern half of the remnant, coincident with ejecta that are part of the bipolar outflow from the remnant. These sites were selected in previous JWST Cycle 1 and NuSTAR Ti-44 emission, but are part of the slowest moving ejecta with low proper motions (<0.2 arcseconds per year), and so will remain in the targeted coordinates and the MIRI/IFU field of view.

We have selected a dedicated background measurement away from the remnant per the requirements for MIRI/MRS observations, which are linked with our two target fields in the observation request.

Proposal 11374 - Targets - A JWST/MIRI Search for Elements Beyond the Iron Peak in Cassiopeia A

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	CASSIOPEIA-A-OUTFLOW-1	RA: 23 23 33.9395 (350.8914146d) Dec: +58 49 0.02 (58.81667d) Equinox: J2000	Proper Motion RA: 135.526 mas/yr Proper Motion Dec: 30.029 mas/yr Parallax: 0.0" Epoch of Position: 2026.395890	
	Comments: Category=ISM Description=[Supernova remnants] Extended=YES				
	(2)	CASSIOPEIA-A-TI44-1	RA: 23 23 32.3377 (350.8847404d) Dec: +58 48 54.64 (58.81518d) Equinox: J2000	Proper Motion RA: +100.341 mas/yr Proper Motion Dec: 14.829 mas/yr Parallax: 0.0" Epoch of Position: 2026.395890	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Supernova remnants] Extended=YES				
	(3)	BACKGROUND	RA: 23 23 47.7100 (350.9487917d) Dec: +58 46 48.30 (58.78008d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES				

Proposal 11374 - Observation 1 - A JWST/MIRI Search for Elements Beyond the Iron Peak in Cassiopeia A

Observation	Proposal 11374, Observation 1: MIRI IFU Ti44 1												Thu May 28 20:00:11 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI IFU Background (Obs 3)]													
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous				
	(2)	CASSIOPEIA-A-TI44-1		RA: 23 23 32.3377 (350.8847404d)			Proper Motion RA: +100.341 mas/yr			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Supernova remnants] Extended=YES				
	Dec: +58 48 54.64 (58.81518d)			Proper Motion Dec: 14.829 mas/yr										
	Equinox: J2000			Parallax: 0.0"										
	Epoch of Position: 2026.395890													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	F560W	Channel 1			YES			FULL			Allow Auto Reorder			
Dithers	#	Dither Type			Optimized For			Direction						
	1	4-Point			EXTENDED SOURCE			NEGATIVE						
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	SLOWR1	12	6	1	Dither 1	4	24	7358.095		
	1	MEDIUM(B)	MRSLONG		SLOWR1	21	4	1	Dither 1	4	16	8313.692		
	1	MEDIUM(B)	MRSSHORT		SLOWR1	21	4	1	Dither 1	4	16	8313.692		

Proposal 11374 - Observation 1 - A JWST/MIRI Search for Elements Beyond the Iron Peak in Cassiopeia A

Special Requirements	Sequence Observations 3, 1, 2 (reordered), Non-interruptible
----------------------	--

Proposal 11374 - Observation 2 - A JWST/MIRI Search for Elements Beyond the Iron Peak in Cassiopeia A

Observation	Proposal 11374, Observation 2: MIRI IFU Outflow 1												Thu May 28 20:00:11 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI IFU Background (Obs 3)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(1)	CASSIOPEIA-A-OUTFLOW-1	RA: 23 23 33.9395 (350.8914146d) Dec: +58 49 0.02 (58.81667d) Equinox: J2000				Proper Motion RA: 135.526 mas/yr Proper Motion Dec: 30.029 mas/yr Parallax: 0.0" Epoch of Position: 2026.395890							
	Comments: Category=ISM Description=[Supernova remnants] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
	F560W	Channel 1				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	SLOWR1	12	6	1	Dither 1	4	24	7358.095		
	1	MEDIUM(B)	MRSLONG		SLOWR1	21	4	1	Dither 1	4	16	8313.692		
	1	MEDIUM(B)	MRSSHORT		SLOWR1	21	4	1	Dither 1	4	16	8313.692		

Proposal 11374 - Observation 2 - A JWST/MIRI Search for Elements Beyond the Iron Peak in Cassiopeia A

Special Requirements	Sequence Observations 3, 1, 2 (reordered), Non-interruptible
----------------------	--

Proposal 11374 - Observation 3 - A JWST/MIRI Search for Elements Beyond the Iron Peak in Cassiopeia A

Observation	Proposal 11374, Observation 3: MIRI IFU Background												Thu May 28 20:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [MIRI IFU Ti44 1 (Obs 1), MIRI IFU Outflow 1 (Obs 2)]												
Comments: Background for Obs 1													
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(3)	BACKGROUND	RA: 23 23 47.7100 (350.9487917d)										
			Dec: +58 46 48.30 (58.78008d)										
			Equinox: J2000										
Acquisition	Comments:												
	Category=Calibration												
	Description=[Telescope/sky background]												
	Extended=YES												
	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	Channel 1				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					BACKGROUND			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F560W	SLOWR1	12	6	1	Dither 1	4	24	7358.095	
	1	MEDIUM(B)	MRSLONG		SLOWR1	21	4	1	Dither 1	4	16	8313.692	
	1	MEDIUM(B)	MRSSHORT		SLOWR1	21	4	1	Dither 1	4	16	8313.692	

Proposal 11374 - Observation 3 - A JWST/MIRI Search for Elements Beyond the Iron Peak in Cassiopeia A

Special Requirements	Sequence Observations 3, 1, 2 (reordered), Non-interruptible
----------------------	--